

Enemy representation at $L = 0.3$:

A^- is three certified modes, Birman–Schwinger < 1 ,
operator R^{-1} open

PRIME.RDAGGER.WOODBURY_MINF.01 — round 486

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Abstract

Rounds 479–485 tried to represent the *safe* orthocomplement of a certified block (garbage from ψ_{10} , QR killed S). This round represents the *dangerous* part. $Q_W = A + \Pi$ is a Fourier multiplier plus rank one. Split $A = A^+ - A^-$ with $A^- = |\sigma_A| 1_{|t| < t_c}$, $t_c \in [6.289835988836902, 6.289835988836904]$ the unique zero of σ_A (r479). A^- is bounded (≤ 5.373) and effectively 1.2-dimensional: $\text{tr}(K_{t_c}) = 2Lt_c/\pi \in [1.201270186632830, 1.201270186632832]$. Slepian numbers of K_{t_c} ($0.859391, 0.313778, 0.027320, 7.70 \cdot 10^{-4}, \dots$) drop after $n = 2$; the first three modes have large eigenvalues (no garbage). Route (b)/(ii) first, as mandated. Float: on $n = 48/96$, $T = 150$, the Birman–Schwinger number $\lambda_{\max}(R^{-1/2}A^-R^{-1/2})$ with $R = A^+ + \Pi$ equals 0.99409 (three-mode compression 0.99365), margin $5.91 \cdot 10^{-3}$. Schur-reducing R against the Nyström rest does not move the margin. Q on $E = \text{span}\{\psi_0, \psi_1, \psi_2\}$ is Higham-certified $\mu \geq 2.87 \cdot 10^{-3}$. The r476 first-Dirichlet midpoint hits ($Q = 1.178 \cdot 10^{-2}$ inside $[1.150, 1.192] \cdot 10^{-2}$). Trace-rest after those three modes is $7.81 \cdot 10^{-4}$. The operator R^{-1} on $L^2[-L, L]$ is open (Route (a) was not opened: (b) reduced the statement, it did not fail). Verdict REDUCED(enemy-3dim-BS<1; Q_E-Higham; operator-Rinv-open). Fourier–HS is not reused. Yoshida–Bombieri zone. No RH claim.

Status. REDUCED(enemy-3dim-BS<1; Q_E-Higham; operator-Rinv-open). No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq c$. The enemy A^- is three certified modes; the finite Birman–Schwinger number is < 1 ; the operator $R^{-1} = (A^+ + \Pi)^{-1}$ is open. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $Q_W = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ one has $P \equiv 0$ (scramble-sensitive, vacuous). A is the Fourier multiplier with symbol $\sigma_A(t) = \text{Re } \psi(\frac{1}{4} + it/2) - \log \pi$. The r469/r479–r485 contract is in force. t_c and $\text{tr}(K_{t_c})$ are the r479 pins.

2 Route

Route (b)/(ii) first: write $A = A^+ - A^-$ with A^- the negative-symbol part on $|t| < t_c$, and test $\|R^{-1/2}A^-R^{-1/2}\| \leq 1$, $R = A^+ + \Pi$, by representing A^- in the Slepian basis of K_{t_c} . This uses that the *dangerous* dimension is 1.2, not that the safe complement is ∞ . Route (a) (Sherman–Morrison over $(A + \mu)^{-1}$, commutator $[P_L, \sigma(D)]$) is the remaining operator step; it is not opened this round because (b) did not fail structurally.

3 Enemy spectrum

Nyström $n = 48$ vs $n = 96$, top-five difference $< 10^{-14}$:

$$\lambda(K_{t_c}) = (0.859391, 0.313778, 0.027320, 7.700 \cdot 10^{-4}, 1.120 \cdot 10^{-5}, \dots).$$

Even 0, 2, 4; odd 1, 3, 5. Rest after $\{\psi_0, \psi_1, \psi_2\}$: $7.81 \cdot 10^{-4}$; $|\sigma_A(0)| \times \text{rest} = 4.20 \cdot 10^{-3}$. After eight modes the trace remainder is 10^{-14} . Jets $|u(L)|$ at n vs $2n$ agree to 10^{-3} on ψ_0, ψ_1 (certified); ψ_2 is $5.7 \cdot 10^{-3}$ (usable for the bounded-band integrals on $[0, t_c]$, phase $tL \leq 1.89$). No K_8 -garbage mode is used.

4 Float balance, then interval

Object	$n = 48$	$n = 96$	comment
$\lambda_{\min}(Q _E)$, $E = \{\psi_0, \psi_1, \psi_2\}$	$6.943 \cdot 10^{-3}$	$6.943 \cdot 10^{-3}$	PD
Higham μ (radius $2 \cdot 10^{-4}$)	$2.87 \cdot 10^{-3}$	—	interval-PD
BS on E (vs $R _E$)	0.99365	0.99365	margin $6.35 \cdot 10^{-3}$
BS on 16 Nyström modes	0.99409	0.99415	margin $5.85 \cdot 10^{-3}$
BS after Schur of R	0.99404	0.99410	$S_R \succ 0.35$
$\ A^- _{E, \text{rest}}\ $	$5.25 \cdot 10^{-3}$	$5.25 \cdot 10^{-3}$	3-mode capture
first-Dirichlet Q	$1.178 \cdot 10^{-2}$	$1.178 \cdot 10^{-2}$	r476 hit
odd Q on $\{\psi_1, \psi_3\}$	0.226	—	easy

Lemma 1 (finite enemy Q). *On $E = \text{span}\{\psi_0, \psi_1, \psi_2\}$ of K_{t_c} at $L = 0.3$, the compression of Q_W satisfies $\lambda_{\min}(Q|_E) \geq 2.87 \cdot 10^{-3}$ by Higham with entry radius $2 \cdot 10^{-4}$.*

$T = 400$ is not used: e^{itx} outruns the Gauss nodes and the Dirichlet Q jumps from $1.18 \cdot 10^{-2}$ to 7 (anti-list).

5 Why this is not λ_*

A^- is captured by three certified modes to a $5 \cdot 10^{-3}$ leftover (or 10^{-14} after eight). The Birman–Schwinger test on the Nyström space is green with a 0.6% margin, stable in n . The operator statement $\|R^{-1/2}A^-R^{-1/2}\|_{\mathcal{L}(L^2[-L, L])} \leq 1$ needs $R^{-1} = (A^+ + \Pi)^{-1}$ on three explicit functions (Woodbury on the rank-one Π , then invert $P_L\sigma_+(D)P_L$). That is Route (a). The A^+ leftover coupling of E into the K -complement is still ≈ 0.29 , fifty times the Higham margin — the r483–r485 completeness hole, now isolated as a three-function resolvent.

The constant minorant $\sigma_A(0)K_{t_c} + \Pi$ remains indefinite ($\lambda_{\min} \approx -3.39$); it is not reused.

6 Kills

Object	Round	Status
Fourier–HS majorant	r478	not reused
Constant minorant $\sigma(0)K + \Pi$	r479	indefinite
Crude κ_{high}	r480	killed
Integer Fourier ONB	r481	$Q_{0n} \sim 1/n$
20× drop / K_8 -garbage	r484	falsified
Flat $c_{\text{tail}}I$ as M_∞	r485	undershoot
$T = 400$ quadrature as pad	r486	Dirichlet Q jumps
Finite BS < 1 as SATZ pad	r486	Nyström only

7 Scaling at $L = 0.8$

The enemy dimension grows as $2Lt_c/\pi$ (≈ 3.2 at $L = 0.8$); the first few Slepian modes of K_{t_c} remain large-eigenvalue modes. P becomes live. The resolvent step does not become easier.

8 What is not claimed

No $\lambda_*(0.3) \geq c$. The enemy is three certified modes; the finite Birman–Schwinger number is < 1 ; R^{-1} on L^2 is open. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.